Gautam Iyer

Assistant Professor

Dept. of Math. Sciences,
Wean Hall #6113,
Carnegie Mellon University Pittsburgh, PA 15213
www.math.cmu.edu/~gautam
Office: 6124 WEH

Education

2006 **Ph.D. in Mathematics**, *University of Chicago*.

2001 **M.S. in Mathematics**, *University of Chicago*.

1999 **B.Sc. in Mathematics**, *St. Xaviers's College, Mumbai*.

Employment

2009–present **Assistant Professor**, *Mathematical Sciences*, Carnegie Mellon University.

2006–2009 **Szegö Assistant Professor**, *Mathematics*, Stanford University.

Grants

2013–2018 **NSF DMS-1252912**, *CAREER: Anomalous Diffusion, Homogenization and Averaging.*.

2010–2013 **NSF DMS-1007914**, *A probabilistic approach to the Navier-Stokes equations*.

2007–2010 **NSF DMS-0707920**, *A stochastic Lagrangian approach to non-linear transport equations*.

Awards and Scholarships

2013 **Alfred P. Sloan Research Fellow**.

2003 **Wayne C. Booth Graduate student prize**, *University of Chicago*.
Excellence in teaching

2003 **Lawrence and Josephine Graves Teaching Prize**, *University of Chicago*.
Best record in effective and responsible teaching

1999–2000 **McCormick Fellowship**, *University of Chicago*.

1999 **H. M. Mehta (Bar-at-law) Memorial Scholarship**.
Highest aggregate in Mathematics in the Third Year B.Sc. exams.

1999 **Bujor Kaikobad Doctor Memorial Prize**.
Highest aggregate in Mathematics and securing a First Class in the Third Year B.Sc exams

Preprints

[1] G. Iyer, R. L. Pego, and N. Leger, *Limit theorems for Smoluchowski dynamics associated with critical continuous-state branching processes*, Ann. Appl. Probab. (2012), submitted, available at `arXiv:1212.6451`.

[2] G. Iyer, A. Novikov, L. Ryzhik, and T. Komorowski, *From homogenization to averaging in cellular flows*, Ann. Inst. H. Poincaré Anal. Non Linéaire (2011), to appear, available at `arXiv:1108.0074`.

Published Articles

[3] G. Iyer and K. C. Zygalakis, *Numerical studies of homogenization under a fast cellular flow*, SIAM Multiscale Model. Simul. **10** (2012), no. 3, DOI 10.1137/120861308.

[4] G. Iyer, R. L. Pego, and A. D. Zarnescu, *Coercivity and stability results for an extended Navier-Stokes system*, J. Math. Phys. **53** (2012), no. 11, 115–605, DOI 10.1063/1.4738637.

[5] P. Constantin and G. Iyer, *A stochastic-Lagrangian approach to the Navier-Stokes equations in domains with boundary*, Ann. Appl. Probab. **21** (2011), no. 4, 1466–1492, DOI 10.1214/10-AAP731. MR2857454

[6] G. Iyer, A. Novikov, L. Ryzhik, and A. Zlatoš, *Exit times of diffusions with incompressible drift*, SIAM J. Math. Anal. **42** (2010), no. 6, 2484–2498, DOI 10.1137/090776895. MR2733257

[7] G. Iyer and A. Novikov, *The regularizing effects of resetting in a particle system for the Burgers equation*, Ann. Probab. **39** (2011), no. 4, 1468–1501, DOI 10.1214/10-AOP586. MR2857247

[8] P. Constantin, G. Iyer, and J. Wu, *Global regularity for a modified critical dissipative quasi-geostrophic equation*, Indiana Univ. Math. J. **57** (2008), no. 6, 2681–2692, DOI 10.1512/iumj.2008.57.3629. MR2482996 (2009k:35120)

[9] G. Iyer and J. Mattingly, *A stochastic-Lagrangian particle system for the Navier-Stokes equations*, Nonlinearity **21** (2008), no. 11, 2537–2553, DOI 10.1088/0951-7715/21/11/004. MR2448230 (2009h:76060)

[10] G. Iyer, *A stochastic Lagrangian proof of global existence of the Navier-Stokes equations for flows with small Reynolds number*, Ann. Inst. H. Poincaré Anal. Non Linéaire **26** (2009), no. 1, 181–189, DOI 10.1016/j.anihpc.2007.10.003. MR2483818

[11] P. Constantin and G. Iyer, *Stochastic Lagrangian transport and generalized relative entropies*, Commun. Math. Sci. **4** (2006), no. 4, 767–777. MR2264819 (2009a:60073)

[12] G. Iyer, *A stochastic Lagrangian formulation of the Navier-Stokes and related transport equations*, Ph. D. Thesis, University of Chicago, 2006.

[13] P. Constantin and G. Iyer, *A stochastic Lagrangian representation of the three-dimensional incompressible Navier-Stokes equations*, Comm. Pure Appl. Math. **61** (2008), no. 3, 330–345, DOI 10.1002/cpa.20192. MR2376844 (2009b:35319)

[14] G. Iyer, *A stochastic perturbation of inviscid flows*, Comm. Math. Phys. **266** (2006), no. 3, 631–645, DOI 10.1007/s00220-006-0058-5. MR2238892 (2007e:76051)

[15] G. Iyer, *Transport in viscous rotating fluids*, Commun. Math. Sci. **2** (2004), no. 4, 673–684. MR2119933 (2005k:76136)

[16] G. Iyer, *Continuous Vector Products in* $\mathbb{R}^n$, The Bombay Mathematics Colloquium Bulletin (1998), [Undergraduate publication].

Professional Activities

Conferences and workshops organized.

Jul, 2013 Mathematics Research Communities: Regularity problems for nonlinear partial differential equations modeling fluids and complex fluids., *Utah.*

Nov, 2011 SIAM Mini symposium, Fluid Dynamics: Deterministic and Stochastic perspectives, *San Diego.*

Oct, 2011 CNA Conference, Incompressible fluids, Turbulence and Mixing, *Carnegie Mellon.*

Editorial work

2012 Guest Associate Editor, *J. Math. Phys., Special Issue: Incompressible Fluids, Turbulence and Mixing.*

Educational Activities

Courses taught

CMU *Undergraduate:* Honors Vector Analysis, Linear Algebra, PDE, Reading and Research. *Graduate:* Measure Theory, Stochastic Calculus. *Working Group's (co-organized):* Fluid dynamics.

Stanford *Undergraduate:* Linear Algebra, Nonlinear Dynamics, Real Analysis, PDE, Measure Theory, Functional Analysis. *Graduate:* PDE.

Chicago *Undergraduate:* Calculus, Honors Calculus, Multi-variable Calculus.